

Untitled model

Introduction to semantic modelling

A **concept scheme** (in SKOS: *ConceptScheme*) is a structured collection of concepts that carry a fixed, shared meaning within a particular domain. The goal is for everyone working with the model — people and systems alike — to use the same words for the same things. That sounds obvious, but in practice language causes a surprising amount of confusion: what one organisation calls a *measurement point*, another calls a *sensor*, a *location*, or an *observation station*.

A concept scheme solves this by recording, for each concept:

- a **preferred label** (prefLabel) it is known by,
- a **definition** that unambiguously describes what the concept does and does not cover,
- and optionally a **scope note** or **example** to clarify borderline cases.

Concepts rarely stand on their own. They are related **hierarchically** (a vehicle is broader than a truck) or **associatively** (a road user relates to a road segment). These relationships make it possible for queries and analyses to navigate between general and specific concepts, and to make connections between concepts explicit. Within SKOS we use `broader`, `narrower`, and `related` for that.

When a concept also exists elsewhere — in a national or European standard, for instance — we record that as a **mapping relation**. This prevents reinventing the wheel: we link to the source so interoperability with external systems remains possible.

This document describes the concept scheme as it is currently recorded. The following sections list the metadata, every concept, the relationships between them, and — where the model is connected to a data source — how it is linked to that source data.

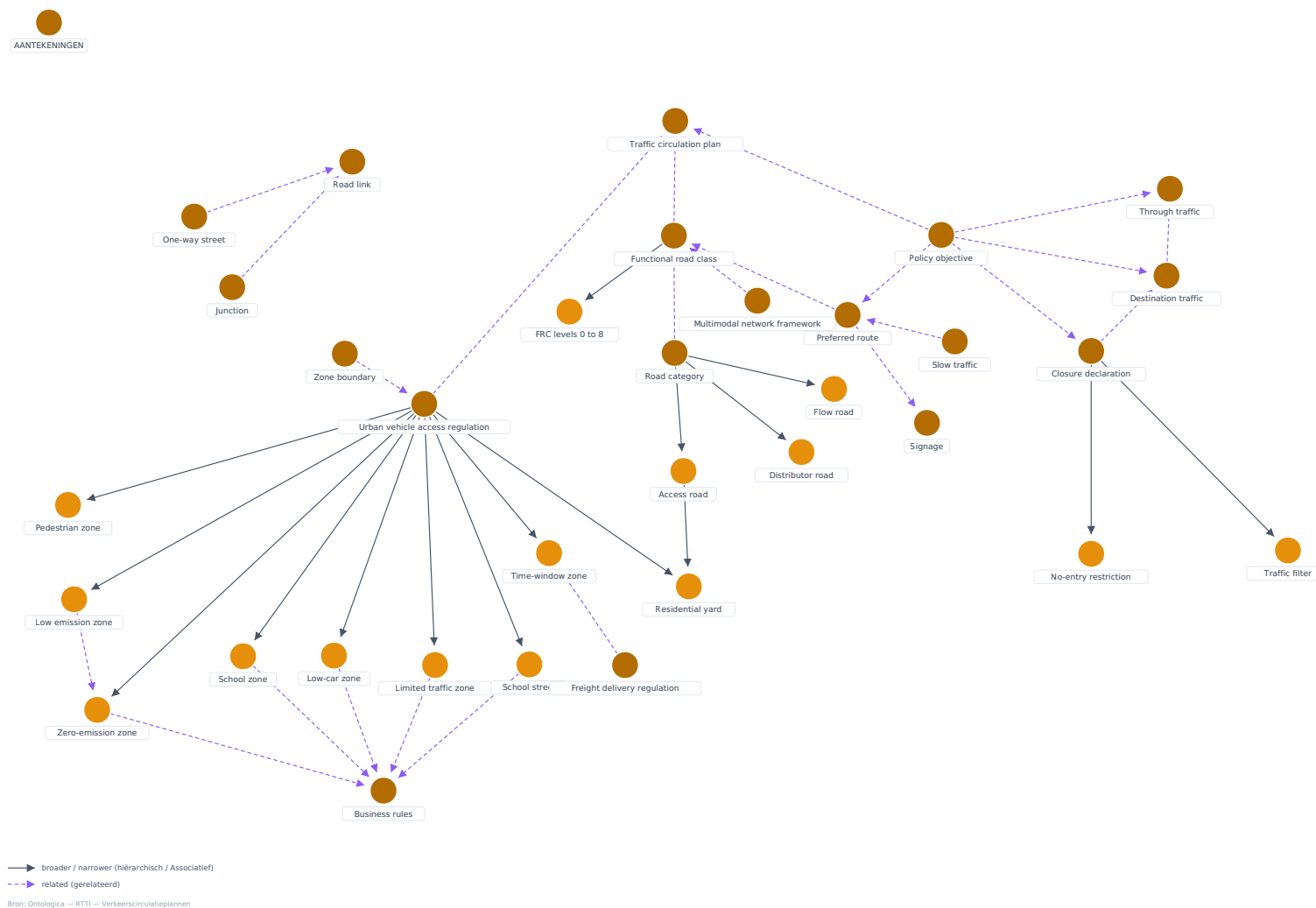
Naamloos model

Metadata

Field	Value
URI	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-verkeerscirculatieplannen
Version	0.5.0
Publication status	Draft
Management status	Active
Organization	Nationaal Toegangspunt Mobiliteit
Created on	2026-03-19
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Last modified by	arthur.keesen@ndw.nu

Model overview

Visual overview of the concepts and their relations. Hierarchical relations are drawn with solid arrows; associative relations are drawn with dashed arrows.



Concepts

Concepts overview

Concept	Definition	Scope	Source
Traffic circulation plan	A coherent set of permanent or recurring traffic management measures designed by a road authority to guide traffic flows within an area, taking into account known and recurring traffic situations, seasonal influences and existing restrictions (e.g. school zones).	A traffic circulation plan does not itself introduce restrictions.	https://eur-lex.europa.eu/eli/reg_del/2022/670/oj , https://www.polisnetwork.eu/wp-content/uploads/2024/11/RTTI-Report.pdf
Policy objective	An objective formulated at policy level - such as accessibility, liveability, road safety or air quality - that guides the desired traffic flows within a traffic circulation plan.	In line with the POLIS/CROW position paper, policy objectives are the reason for the choice of measures and preferred routes in a traffic circulation plan.	https://www.polisnetwork.eu/wp-content/uploads/2024/11/RTTI-Report.pdf
Multimodal network framework	An unambiguous representation of the multimodal use of the road network desired as a matter of policy, with a prioritisation between modes per network element.	Standardised through the CROW/LVMB guideline and manual for multimodal network frameworks. Forms the modality layer to which preferred routes and road categorisation can connect.	https://www.crow.nl/kennisproducten/richtlijn-en-leidraad-voor-multimodale-netwerkkaders/
Functional road class	A classification of roads based on the importance of their function within the road network, independent of the physical road category.	Based on the INSPIRE Transport Networks FunctionalRoadClass. The POLIS/CROW position paper proposes the functional road classification as the basis (data type 2b) for the digital traffic circulation plan, with the caveat that policy purposes often require more	https://inspire.ec.europa.eu/enumeration/FunctionalRoadClassValue , https://www.polisnetwork.eu/wp-content/uploads/2024/11/RTTI-Report.pdf

Concept	Definition	Scope	Source
		functions than the European FRC offers and that multimodality is still lacking.	
FRC levels 0 to 8	The nine levels of the functional road classification, ranging from level 0 (main road/motorway) up to level 8 (local roads of minor importance and paths).	Value list in line with the INSPIRE FunctionalRoadClassValue enumeration, as presented in the POLIS/CROW position paper. The individual level URIs follow the pattern of the INSPIRE code list register.	https://inspire.ec.europa.eu/enumeration/FunctionalRoadClassValue , https://www.polisnetwork.eu/wp-content/uploads/2024/11/RTTI-Report.pdf
Road category	The physical-functional classification of a road according to the Dutch road design system, determining its layout and intended use.	Based on the Sustainable Safety (Duurzaam Veilig) categorisation (CROW). The Dutch road categories correspond functionally to the European functional road classification, but do not form a one-to-one hierarchy with it; hence a skos:related relation instead of broader/narrower.	https://www.crow.nl/Onderwerpen/wegontwerp-en-weginrichting/duurzaam-veilig-wegontwerp/
Flow road	A road category aimed at the through-flow of traffic at a constant, generally high speed, with grade-separated intersections.	Sustainable Safety (Duurzaam Veilig) category; occurs mainly outside built-up areas.	https://www.crow.nl/Onderwerpen/wegontwerp-en-weginrichting/duurzaam-veilig-wegontwerp/
Distributor road	A road category that provides access to rural or urban areas: through-flow on the links, exchange at the intersections.	Sustainable Safety (Duurzaam Veilig) category (CROW thesaurus). Abbreviated GOW.	https://data.crow.nl/thesaurus/term/8504D234-DD69-4A8D-BA4E-E207E57ECEAD , https://www.crow.nl/Onderwerpen/wegontwerp-en-weginrichting/duurzaam-veilig-wegontwerp/
Access road	A road category aimed at providing access to premises and destinations, with a residential function and a low speed.	Sustainable Safety (Duurzaam Veilig) category (CROW). Abbreviated ETW.	https://www.crow.nl/Onderwerpen/wegontwerp-en-weginrichting/duurzaam-veilig-wegontwerp/
Residential yard	A residential area designated as a yard (Dutch: erf), bounded by signs G5 (start) and G6 (end), where pedestrians	RVV 1990 (signs G5/G6). A woonerf is the most residential-oriented elaboration of an access road.	https://wetten.overheid.nl/BWBR0004825

Concept	Definition	Scope	Source
	may use the full width of the road and vehicles travel at walking pace (15 km/h).		
Through traffic	Traffic that has neither an origin nor a destination within the area under consideration.	Term in line with the CROW thesaurus. Keeping through traffic out of residential areas is a core objective of many traffic circulation plans.	https://data.crow.nl/thesaurus/term/48FA2203-699F-48A2-B82F-A0C4F6CA181C
Destination traffic	Drivers whose journey destination concerns one or more plots situated on or near a road subject to a closure declaration and reachable for that purpose solely via that road, as well as drivers of scheduled buses.	Definition derived from art. 1 RVV 1990. Destination traffic is usually exempted from a closure declaration by means of a supplementary panel.	https://wetten.overheid.nl/BWBR0004825
Preferred route	A route desired by the road authority as a matter of policy, per mode or target group, intended to be shared with service providers so that navigation services can guide traffic along it.	One of the use cases of a digital traffic circulation plan mentioned in the POLIS/CROW position paper (alongside areas to be avoided and the use of main routes).	https://www.polisnetwork.eu/wp-content/uploads/2024/11/RTTI-Report.pdf
Slow traffic	Non-motorised vehicles, cyclists, pedestrians and persons leading or riding an animal, as well as motor vehicles that can or may travel only at a low speed.	Term in line with the CROW thesaurus. Relevant because circulation measures usually exempt slow traffic (e.g. a traffic filter that cyclists can still pass).	https://data.crow.nl/thesaurus/term/264DF64F-903B-49CD-A4E4-D3E235A20FF2
Urban vehicle access regulation	A limitation or restriction on the use of the road network on the basis of vehicle type, traveller type or time, applying to a delimited area.	Known by the European term UVAR (Urban Vehicle Access Regulation). Placed in DATEX II in the TrafficRegulation and ControlledZone namespaces; the NAPCORE Quality Framework for UVAR Data describes the associated data quality requirements.	https://datex2.eu/user-domains/uvars/ , https://napcore.eu/quality/

Concept	Definition	Scope	Source
Low emission zone	A zone from which motor vehicles that do not meet a given emission class are barred, in order to improve air quality; known across Europe as a low emission zone (LEZ).	Indicated by sign C22a, which together with the other C22 signs is being replaced by the unified sign C22e (Stb. 2025, 265).	https://wetten.overheid.nl/BWBR0004825 , https://zoek.officielebekendmakingen.nl/stb-2025-265.html
Zero-emission zone	A zone where only zero-emission vehicles are permitted for vans and lorries.	Since 1 January 2025 municipalities have been introducing zero-emission zones for urban logistics. Indicated by sign C22c, which is being merged into the unified sign C22e (Stb. 2025, 265).	https://wetten.overheid.nl/BWBR0004825 , https://zoek.officielebekendmakingen.nl/stb-2025-265.html
Pedestrian zone	An area in which, in principle, only pedestrians may be present and where they have priority over other traffic.	Also referred to as a pedestrian area; indicated by signs G7/G8 (RVV 1990).	https://wetten.overheid.nl/BWBR0004825
Low-car zone	An area with deliberately limited motorised traffic, usually realised through exemptions and licence plate enforcement.	Not defined as such in the RVV; in practice a combination of closure declarations, time windows and exemptions.	https://wetten.overheid.nl/BWBR0004825
Time-window zone	An area accessible to (delivery) motorised traffic only at times determined by the municipality.	Time windows are made known by means of supplementary panels accompanying a closure declaration (BABW). Widely used for urban logistics.	https://wetten.overheid.nl/BWBR0004826
School street	A street that is car-free during the start and end times of the school day.	Mentioned as a concrete measure in the POLIS/CROW position paper; in the Netherlands usually realised through a time-bound closure declaration.	https://www.polisnetwork.eu/wp-content/uploads/2024/11/RTTI-Report.pdf
Limited traffic zone	A delimited area within which traffic is restricted for certain vehicle categories or target groups; known internationally as ZTL (zona a traffico limitato).	Generic zonal regime covering international UVAR practice, shaped in the Netherlands through a closure declaration with exemptions.	https://datex2.eu/user-domains/uvars/

Concept	Definition	Scope	Source
One-way street	A road link where traffic is permitted in one direction of travel only.	Indicated by signs C2/C3/C4 (RVV 1990); to be modelled in DATEX II as a TrafficRegulation. One-way traffic is a widely used circulation measure.	https://docs.datex2.eu/levels/mastering/trafficregulation/ , https://wetten.overheid.nl/BWBR0004825
Closure declaration	A prohibition on entering and using the road or road section concerned.	Definition derived from art. 1 RVV 1990; realised through the C series of signs. Forms the legal basis for many circulation measures.	https://wetten.overheid.nl/BWBR0004825
No-entry restriction	A permanent access restriction for one or more vehicle categories on a specific road link.	Falls under the RTTI data category permanent access restrictions (NAPCORE data dictionary).	https://napcore.eu/wp-content/uploads/2025/05/DD_2022_670.pdf
Traffic filter	A physical closure - for example by means of retractable bollards - that makes through motorised traffic impossible, while cyclists, pedestrians and exemption holders can pass.	The traffic filter is the physical realisation of a closure declaration and a core measure against rat-running traffic in traffic circulation plans.	https://wetten.overheid.nl/BWBR0004825
Freight delivery regulation	Regulations for the loading and unloading of goods, such as the designation of road links or areas and time-bound restrictions.	Falls under the RTTI data category freight delivery regulations (NAPCORE data dictionary); closely linked to time windows.	https://napcore.eu/wp-content/uploads/2025/05/DD_2022_670.pdf
Zone boundary	The spatial delimitation of an area within which a zonal traffic rule applies.	Falls under the RTTI data category boundaries of zonal restrictions, prohibitions or obligations (NAPCORE data dictionary).	https://napcore.eu/wp-content/uploads/2025/05/DD_2022_670.pdf
Road link	A part of the road network between two consecutive positions, serving as the smallest unit for recording road characteristics and measures.	Recorded in the Netherlands in the National Road Database (NWB) and as Wegdeel in the BGT/IMGeo. These registers serve here solely as a reference source (dct:source), not as a linking target.	https://docs.geostandaarden.nl/imgeo/catalogus/imgeo/ , https://napcore.eu/wp-content/uploads/2025/05/DD_2022_670.pdf , https://www.nationaalgeoregister.nl/geonetwork/srv/api/records/a9b7026e-0a81-4813-93bd-ba49e6f28502

Concept	Definition	Scope	Source
Junction	A location where traffic can be directed onto another route.	Falls under the RTTI data category junctions (NAPCORE data dictionary). For the geometric realisation, NWB, BGT/IMGeo and the Rotterdam Mercator/IBRO model serve as reference sources (dct:source).	https://docs.geostandaarden.nl/ibro/scope/ , https://napcore.eu/wp-content/uploads/2025/05/DD_2022_670.pdf , https://www.nationaalgeoregister.nl/geonetwork/srv/api/records/a9b7026e-0a81-4813-93bd-ba49e6f28502
Business rules	to be decided later		
School zone	to be decided later		
AANTEKENINGEN	Digital indicators?		
Signage	t.b.d		

Relations per concept

Outgoing relations only. Semantic relations stay inside the model; mapping relations link to concepts elsewhere.

Policy objective

Semantic relations

Type	Target label	Target concept
related	Destination traffic	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000015
related	Traffic circulation plan	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000001
related	Closure declaration	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000028
related	Through traffic	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000014
related	Preferred route	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000017

Multimodal network framework

Semantic relations

Type	Target label	Target concept
related	Functional road class	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000006

Functional road class

Semantic relations

Type	Target label	Target concept
narrower	FRC levels 0 to 8	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000007
related	Traffic circulation plan	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000001

Mapping relations

Type	Target URI
exactMatch	https://docs.datex2.eu/recommended-profiles/rrp/rtti/rtti-670/1-i-b-road-classification/

FRC levels 0 to 8

Semantic relations

Type	Target label	Target concept
broader	Functional road class	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000006

Road category

Semantic relations

Type	Target label	Target concept
narrower	Flow road	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000009
narrower	Distributor road	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000010
narrower	Access road	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000012
related	Functional road class	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000006

Flow road

Semantic relations

Type	Target label	Target concept
broader	Road category	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000008

Distributor road

Semantic relations

Type	Target label	Target concept
broader	Road category	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000008

Access road

Semantic relations

Type	Target label	Target concept
broader	Road category	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000008
narrower	Residential yard	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000013

Residential yard

Semantic relations

Type	Target label	Target concept
broader	Urban vehicle access regulation	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000019

Type	Target label	Target concept
broader	Access road	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000012

Destination traffic

Semantic relations

Type	Target label	Target concept
related	Through traffic	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000014

Preferred route

Semantic relations

Type	Target label	Target concept
related	Signage	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-verkeerscirculatieplannen/concept-1787733042482
related	Functional road class	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000006

Slow traffic

Semantic relations

Type	Target label	Target concept
related	Preferred route	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000017

Urban vehicle access regulation

Semantic relations

Type	Target label	Target concept
narrower	Residential yard	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000013
narrower	Limited traffic zone	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000026
narrower	School street	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000025
narrower	Time-window zone	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000024
narrower	Low-car zone	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000023
narrower	Pedestrian zone	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000022
narrower	Zero-emission zone	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000021
narrower	Low emission zone	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000020
narrower	School zone	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-verkeerscirculatieplannen/concept-1784202761035
related	Traffic circulation plan	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000001

Low emission zone

Semantic relations

Type	Target label	Target concept
broader	Urban vehicle access regulation	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000019
related	Zero-emission zone	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000021

Zero-emission zone

Semantic relations

Type	Target label	Target concept
broader	Urban vehicle access regulation	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000019
related	Business rules	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-verkeerscirculatieplannen/concept-1784202276603

Pedestrian zone

Semantic relations

Type	Target label	Target concept
broader	Urban vehicle access regulation	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000019

Low-car zone

Semantic relations

Type	Target label	Target concept
broader	Urban vehicle access regulation	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000019
related	Business rules	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-verkeerscirculatieplannen/concept-1784202276603

Time-window zone

Semantic relations

Type	Target label	Target concept
broader	Urban vehicle access regulation	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000019

School street

Semantic relations

Type	Target label	Target concept
broader	Urban vehicle access regulation	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000019
related	Business rules	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-verkeerscirculatieplannen/concept-1784202276603

Limited traffic zone

Semantic relations

Type	Target label	Target concept
broader	Urban vehicle access regulation	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000019
related	Business rules	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-verkeerscirculatieplannen/concept-1784202276603

One-way street

Semantic relations

Type	Target label	Target concept
related	Road link	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000033

Closure declaration

Semantic relations

Type	Target label	Target concept
narrower	Traffic filter	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000030
narrower	No-entry restriction	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000029
related	Destination traffic	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000015

No-entry restriction

Semantic relations

Type	Target label	Target concept
broader	Closure declaration	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000028

Traffic filter

Semantic relations

Type	Target label	Target concept
broader	Closure declaration	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000028

Freight delivery regulation

Semantic relations

Type	Target label	Target concept
related	Time-window zone	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000024

Zone boundary

Semantic relations

Type	Target label	Target concept
related	Urban vehicle access regulation	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000019

Junction

Semantic relations

Type	Target label	Target concept
related	Road link	https://gegevenscatalogus.gov/rtti-verkeerscirculatieplannen/concept-1773240000033

School zone

Semantic relations

Type	Target label	Target concept
related	Business rules	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-verkeerscirculatieplannen/concept-1784202276603

Introduction to RML mappings

An **information description** (technically: an RML mapping) describes how concrete data — records in a file, rows in a table, fields in an XML message — corresponds to the concepts in the concept scheme above. Without such a description data stands on its own; with it, every piece of data is connected to an unambiguous concept.

The principle is simple: for each data source we define one or more **TriplesMaps**. Each TriplesMap is a recipe that says *"each record in this data source represents an instance of that concept, and these fields are its properties"*. For each field we then specify which concept describes the property (for example `hasStatus`, `liesOnRoadSegment`) and how the field should be read from the source.

The essence for the reader of this document:

- the **TriplesMap table** below shows which data sources are linked to this model,
- the **subject template** shows how a unique URI is constructed for each record,
- the **number of predicates** gives a rough indication of how rich the mapping per source is.

The actual mapping rules (field → concept) are managed inside the Ontologica application itself and can be inspected and edited through the mapping wizard.

Information description

TriplesMaps

Name	Datasource	Subject template	rdf:type	# predicates
values	trafficspeed.xml	https://example.org/values/{value/@lang}	—	1